Additional Task
Perceptron Learning Algorithm for AND, OR, and NOT Gates (K4)

Objective

A Single Layer Perceptron was coded from scratch in Python to learn the AND, OR, and NOT logic gates. For each gate, the weights and bias were printed after every single weight update, and the decision boundary was plotted after every update.

Training Log

AND gate:

```
Epoch 1, sample [0 0], target 0, pred 1 -> weights=[0. 0.], bias=-0.10
Epoch 1, sample [1 1], target 1, pred 0 -> weights=[0.1 0.1], bias=0.00
Epoch 2, sample [0 0], target 0, pred 1 -> weights=[0.1 0.1], bias=-0.10
Epoch 2, sample [0 1], target 0, pred 1 -> weights=[0.1 0. ], bias=-0.20
Epoch 2, sample [1 1], target 1, pred 0 -> weights=[0.2 0.1], bias=-0.10
Epoch 3, sample [0 1], target 0, pred 1 -> weights=[0.2 0. ], bias=-0.20
Epoch 3, sample [1 0], target 0, pred 1 -> weights=[0.1 0. ], bias=-0.30
Epoch 3, sample [1 1], target 1, pred 0 -> weights=[0.2 0.1], bias=-0.20
Converged after epoch 4, no errors left.
```

OR gate:

```
Epoch 1, sample [0 0], target 0, pred 1 -> weights=[0. 0.], bias=-0.10
Epoch 1, sample [0 1], target 1, pred 0 -> weights=[0.  0.1], bias=0.00
Epoch 2, sample [0 0], target 0, pred 1 -> weights=[0.  0.1], bias=-0.10
Epoch 2, sample [1 0], target 1, pred 0 -> weights=[0.1 0.1], bias=0.00
Epoch 3, sample [0 0], target 0, pred 1 -> weights=[0.1 0.1], bias=-0.10
Converged after epoch 4, no errors left.
```

NOT gate:

```
Epoch 1, sample [1], target 0, pred 1 -> weights=[-0.1], bias=-0.10
Epoch 2, sample [0], target 1, pred 0 -> weights=[-0.1], bias=0.00
Converged after epoch 3, no errors left.
```

Decision Boundary Plots

Observation: AND converges in 8 updates, OR in 5 updates, and NOT in 2 updates. All three gates are linearly separable, so the perceptron finds a boundary that classifies every sample correctly, unlike the XOR case, which never converges.

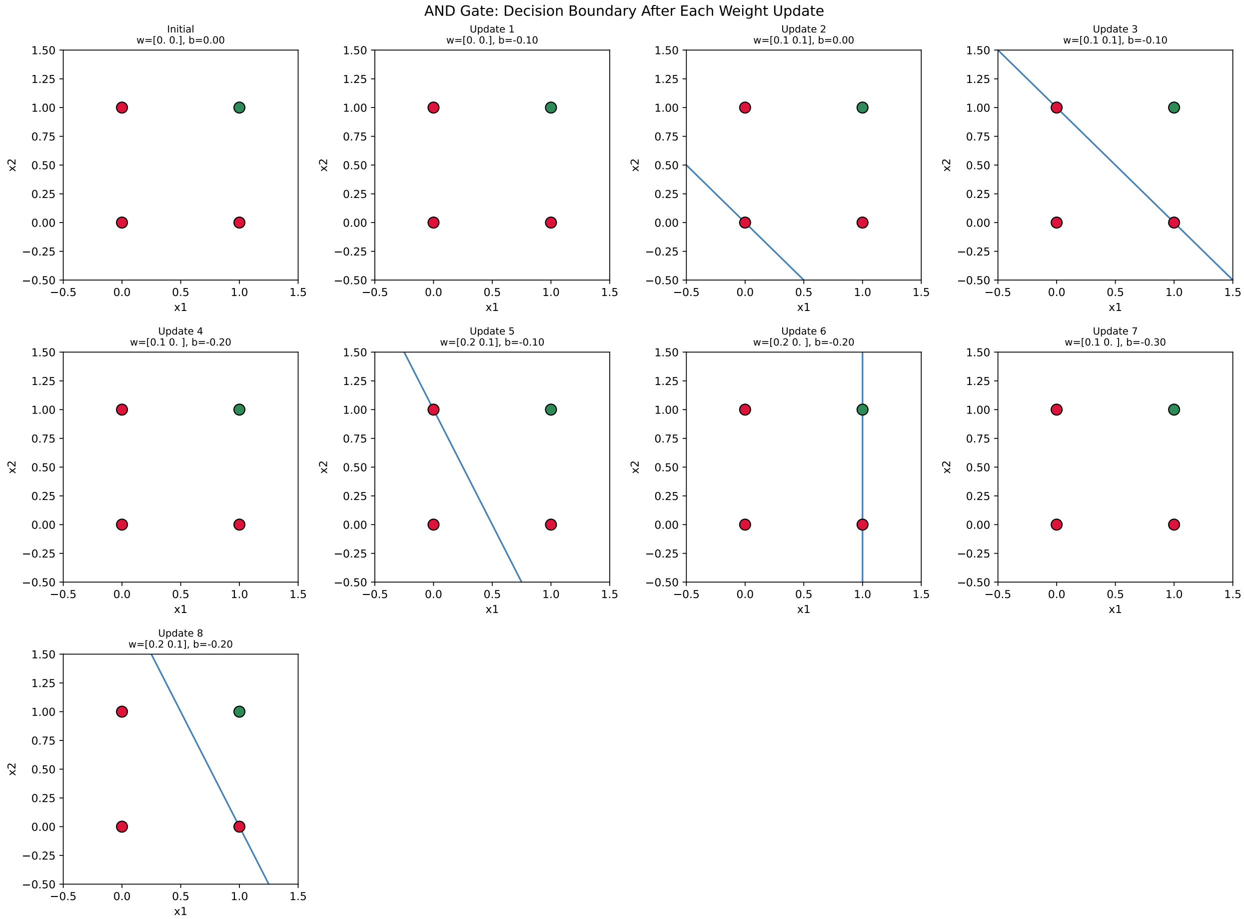

Figure 1: AND Gate: Decision Boundary After Each Weight Update.

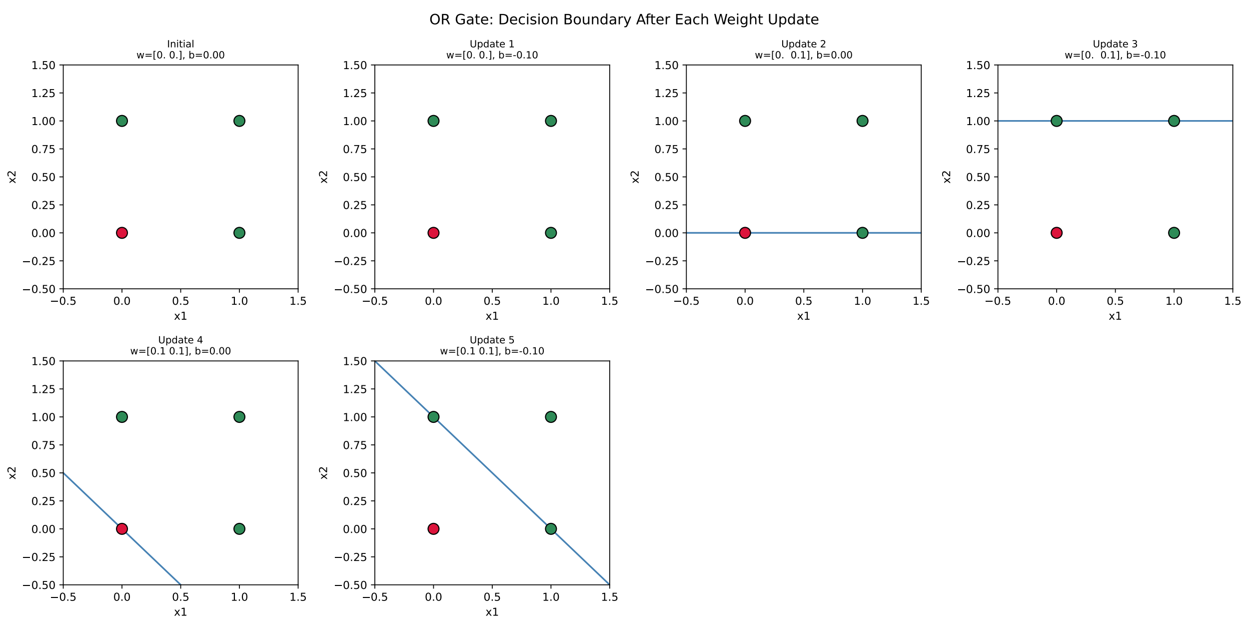

Figure 2: OR Gate: Decision Boundary After Each Weight Update.

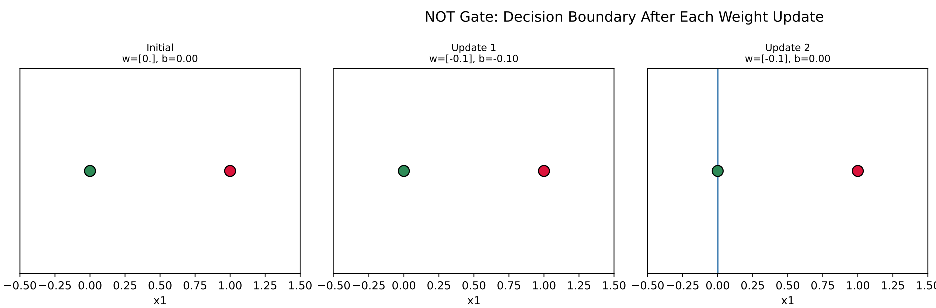

Figure 3: NOT Gate: Decision Boundary After Each Weight Update.